ϕCDM Model

An Introduction to Scalar Field Dark Energy Model

Reference: [astro-ph/0603057], [astro-ph/0803.0982], [astro-ph/0908.4597], [astro-ph/0912.3988]

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April 22, 2010

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2. ΛCDM Model
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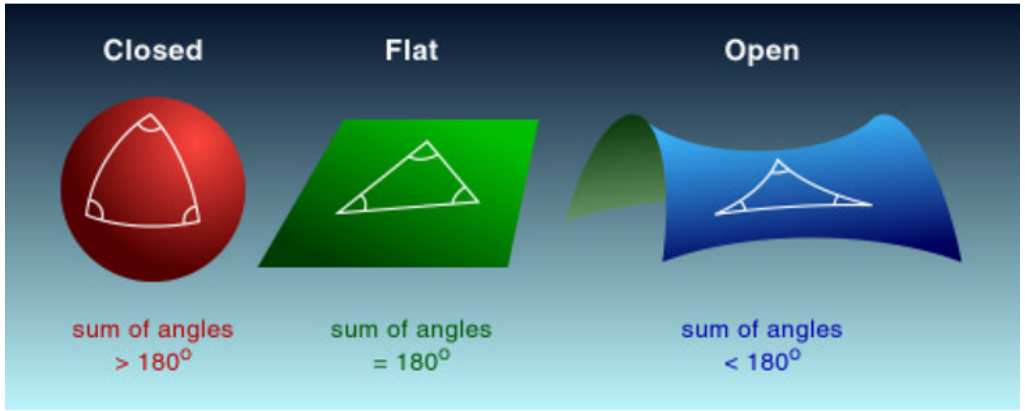
- In Friedmann-Robertson-Walker background (homogeneous and isotropic 3-dim space)

  $\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu = -\mathrm{d}t^2 + a^2(t)\gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j$ ($a(t)$ is a scale factor).

- In spherical coordinates

$$\gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j = \frac{\mathrm{d}r^2}{1-Kr^2} + r^2(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2),$$

$$\Rightarrow \begin{cases} \text{closed}, & K > 0, \\ \text{flat}, & K = 0, \\ \text{open}, & K < 0. \end{cases}$$



http://www.lcsd.gov.hk/

- Einstein equation

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu}.$$

- Energy-momentum tensor for isotropic perfect fluid

$$T^{\mu}{}_{\nu} = \begin{pmatrix} -\rho(t) & 0 & 0 & 0 \\ 0 & P(t) & 0 & 0 \\ 0 & 0 & P(t) & 0 \\ 0 & 0 & 0 & P(t) \end{pmatrix}.$$

- Continuity equation $T^{\mu}{}_{\nu;\mu} = 0$

$$\dot{\rho} = -3(\rho + P)H \Rightarrow \dot{\rho} = -3(1+w)\rho\frac{\dot{a}}{a},$$

with $H \equiv \frac{\dot{a}}{a}$ (Hubble parameter) and $w \equiv \frac{P}{\rho}$ (equation of state),

$$\Rightarrow \rho \propto (a_0/a)^{3(1+w)} \text{ and } a_0 = 1.$$

- The equation of state (EoS)
  - For relativistic particles: $w_{\rm r} = 1/3 \Rightarrow \rho_{\rm r} \propto (a_0/a)^4$.
  - For non-relativistic matter: $w_{\rm m} \simeq 0 \Rightarrow \rho_{\rm m} \propto (a_0/a)^3$.
- Friedmann equation (00 component of Einstein equation)

$$\left(\frac{\dot{a}}{a}\right)^2 = H^2 = \frac{8\pi G}{3}\rho - \frac{K}{a^2},$$

- For $K = 0$, we have critical density $\rho_c \equiv \frac{3H^2}{8\pi G}$ and $\rho_c^{(0)} \equiv \frac{3H_0^2}{8\pi G}$

$$\Rightarrow H = H_0\sqrt{\frac{8\pi G}{3H_0^2}\rho - \frac{K}{a^2 H_0^2}} \text{ with } \Omega_{\rm K} \equiv -\frac{K}{a_0^2 H_0^2},$$

$$\rho = \rho_c^{(0)}\left[\Omega_{\rm m}\left(\frac{a_0}{a}\right)^3 + \Omega_{\rm r}\left(\frac{a_0}{a}\right)^4\right], \text{ and } \Omega_i \equiv \frac{\rho_i^{(0)}}{\rho_c^{(0)}},$$

$$\Rightarrow H = H_0\sqrt{\Omega_{\rm m}\left(\frac{a_0}{a}\right)^3 + \Omega_{\rm r}\left(\frac{a_0}{a}\right)^4 + \Omega_{\rm K}\left(\frac{a_0}{a}\right)^2}.$$

- Acceleration is defined to mean that $\ddot{a} > 0$.

- From ii component Einstein equation

$$\begin{aligned} & \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P) > 0, \\ \Rightarrow \quad & \rho + 3P < 0, \\ \Rightarrow \quad & w < -\frac{1}{3}, \\ \Rightarrow \quad & \text{negative pressure } P < -\frac{\rho}{3}. \end{aligned}$$

- Two possibilities
  - There exists a new component of the Universe with large negative pressure so called dark energy.
  - GR breaks down on cosmological scale must be replaced with a more complete theory of gravity.

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- It is possible to add the cosmological constant term to the Einstein equation.
- The Friedmann equation

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \Rightarrow \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{K}{a^2},$$

$$\Rightarrow H = H_0 \sqrt{\Omega_{\mathrm{m}} \left(\frac{a_0}{a}\right)^3 + \Omega_{\mathrm{r}} \left(\frac{a_0}{a}\right)^4 + \Omega_\Lambda + \Omega_{\mathrm{K}} \left(\frac{a_0}{a}\right)^2}.$$

- ΛCDM model (Λ + standard CDM model)
  - Choose that $\Omega_\Lambda \simeq$0.7, $\Omega_{\mathrm{m}} \simeq$0.25, $\Omega_{\mathrm{b}} \simeq$0.05 and $\Omega_{\mathrm{r}} \simeq$0.

$$\Rightarrow \Omega_{\mathrm{m}} + \Omega_{\mathrm{b}} \equiv \Omega_{\mathrm{M}} \simeq 0.3 \Rightarrow \Omega_{\mathrm{M}} + \Omega_\Lambda = 1 \text{ and } w_\Lambda = -1.$$

  - Corresponding to a flat spatial geometry

$$H = H_0 \sqrt{\Omega_{\mathrm{M}} \left(\frac{a_0}{a}\right)^3 + \Omega_\Lambda}.$$

- Two problems of ΛCDM
  - "Fine Tuning" problem:

$$\Lambda \approx H_0^2 = (2.13h \times 10^{-42}\mathrm{GeV})^2 \Rightarrow \rho_\Lambda = \frac{\Lambda m_{\mathrm{pl}}^2}{8\pi} \approx 10^{-47}\mathrm{GeV}^4 \simeq \rho_c.$$

In QFT, the vaccum energy of scalar field with mass m

$$\begin{aligned} \rho_{\mathrm{vac}} &= \frac{1}{2}\int_0^\infty \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\sqrt{k^2+m^2} \\ &= \frac{1}{4\pi^2}\int_0^\infty \mathrm{d}k k^2 \sqrt{k^2+m^2} \approx \frac{k_{\mathrm{max}}^4}{16\pi^2}. \end{aligned}$$

GR is valid up to th Planck scale $m_{\mathrm{pl}} = 1.22 \times 10^{19}\mathrm{GeV}$

$$\rho_{\mathrm{vac}} \simeq 10^{74}\mathrm{GeV}^4 \Rightarrow \frac{\rho_{\mathrm{vac}}}{\rho_\Lambda} = 10^{121}.$$

  - "Coincidence" problem:
    There is a short period of time when the energy density of matter and cosmological constant are comparible, i.e., the same order in ρ_{M} and ρ_Λ.

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- Modified matter
  - Quintessence (slowly rolling scalar field),
  - K-essence,
  - Perfect fluid models.
- Modified gravity
  - $f(R)$ gravity (modified ΛCDM $\Rightarrow$ non-linear Lagrangian density in terms of R),
  - Scalar-tensor theories (R couples to ϕ with the form: $F(\phi)R$),
  - Braneworld models.
- Inhomogeneous Universe model.

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- Quintessence: a canonical scalar field ϕ with a potential $V(\phi)$ responsible for the late-time cosmic acceleration.
- Assume that a self-interaction scalar field ϕ minimally coupled to gravity is present on cosmological scale.
- The scalar field is described by the Lagrangian density

$$\mathcal{L}_\phi = \sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right].$$

- The equation of motion is

$$\partial_\mu\left(\frac{\delta\mathcal{L}_\phi}{\delta(\partial_\mu\phi)}\right) - \frac{\delta\mathcal{L}_\phi}{\delta\phi} = 0,$$

$$\Rightarrow -\Box\phi + V'(\phi) = 0.$$

- If the scalar field is close to spatially uniform

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0.$$

- The action of scalar field is

$$\begin{aligned} S &= S_{\text{E-H}} + S_\phi + S_{\text{m}} \\ &= \int \mathrm{d}^4x \left[\frac{1}{16\pi G}\sqrt{-g}R + \mathcal{L}_\phi \right] + S_{\text{m}}. \end{aligned}$$

- The energy momentum tensor of the scalar field is given by

$$T^{(\phi)}_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta \mathcal{L}_\phi}{\delta g^{\mu\nu}} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + V(\phi)\right],$$

$$\begin{cases} \rho_\phi = -T^{0(\phi)}_{\ \ 0} = \frac{1}{2}\dot{\phi}^2 + V(\phi) \\ P_\phi = T^{i(\phi)}_{\ \ i} = \frac{1}{2}\dot{\phi}^2 - V(\phi) \end{cases} \Rightarrow w_\phi = \frac{\dot{\phi}^2 - 2V(\phi)}{\dot{\phi}^2 + 2V(\phi)},$$

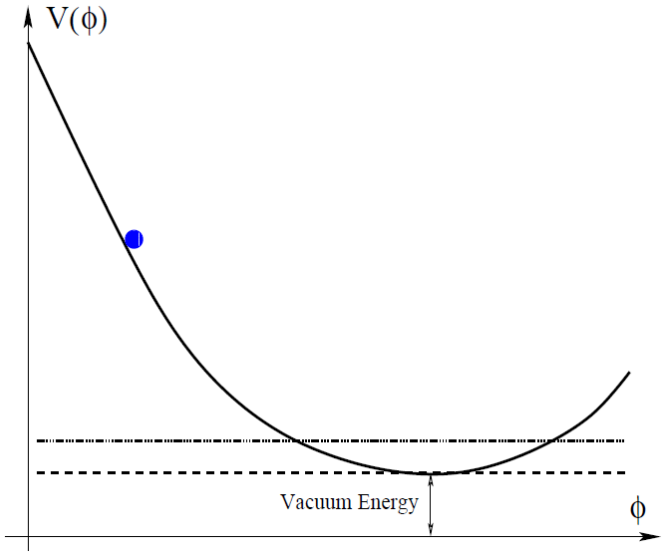
$$H^2 = \frac{8\pi G}{3}\left[\frac{1}{2}\dot{\phi}^2 + V(\phi)\right],$$

$$\frac{\ddot{a}}{a} = -\frac{8\pi G}{3}\left[\dot{\phi}^2 - V(\phi)\right].$$

- If the scalar field evolves slowly $(\dot{\phi}^2 \ll V(\phi))$

$$w_\phi = \frac{P_\phi}{\rho_\phi} \simeq \frac{-2V}{2V} = -1 \;\Rightarrow w_\phi \simeq w_\Lambda$$

- The introduced scalar field behaves like a slowly varying vacuum energy, $\rho_{\rm vac} \simeq V[\phi(t)]$.
  it can be regard as a dynamical vacuum energy effectively.



[astro-ph/0803.0982]

- The Hubble parameter should be rewritten as

$$H = H_0\sqrt{\Omega_{\rm m}\left(\frac{a_0}{a}\right)^3 + \Omega_{\rm r}\left(\frac{a_0}{a}\right)^4 + \Omega_{\rm K}\left(\frac{a_0}{a}\right)^2 + \Omega_\phi}.$$

- We got Ω_ϕ is the time-dependent energy density parameter of the scalar field
- Two specific cases:
  - A self-interaction and inverse-power law potential

$$V(\phi) = M^{4+\alpha}\phi^{-\alpha},$$

  - A exponential potential

$$V(\phi) = m^4 e^{-\lambda\phi/M_{\rm pl}}.$$

- In these two kinds of potential, we require the condition $w_\phi < -\frac{1}{3}$ to realize the late-time cosmic acceleration, which translates into the condition $\dot{\phi}^2 < V(\phi)$.
- The scalar potential needs to be shallow enough for the field to evolve slowly along the potential (slowly rolling scalar field).

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- The method of maximum likelihood - likelihood function
  - A set of N measure quantities $\boldsymbol{x} = (x_1, ..., x_N)$ describe by a joint p.d.f. $f(\boldsymbol{x}; \boldsymbol{\theta})$, where $\boldsymbol{\theta} = (\theta_1, ..., \theta_n)$ is a set of n parameters whose value are unknown.
  - The *likelihood function* $\mathcal{L}(\boldsymbol{\theta}) \equiv f(\boldsymbol{x}; \boldsymbol{\theta})$.
  - If the measurements x_i are statistically independent and each follow the p.d.f. $f(x_i; \boldsymbol{\theta})$, then the joint p.d.f. for x factorizes and the likelihood function is

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{N} f(x_i; \boldsymbol{\theta}) \ .$$

  - It is usually easier to work with $\ln \mathcal{L}$, and since both are maximized for the same parameter value $\boldsymbol{\theta}$, the maximum likelihood (ML) estimators can be found by solving the *likelihood equations*,

$$\frac{\partial \ln \mathcal{L}}{\partial \theta_i} = 0 \ , \qquad i = 1, ..., n \ .$$

- The method of least squares - χ^2 function
  - The *method of least squares* (LS) coincides with the method of maximum likelihood in the following special case.
  - Consider a set of N independent measurements y_i at known points x_i. The measurement y_i is assumed to be Gaussian distributed with mean $F(x_i;\boldsymbol{\theta})$ and known variance σ_i^2.
  - The goal is to construct estimators for the unknown parameters $\boldsymbol{\theta}$,

$$\chi^2(\boldsymbol{\theta}) = -2\ln\mathcal{L}(\boldsymbol{\theta}) + \text{constant} = \sum_{i=1}^{N} \frac{(y_i - F(x_i;\boldsymbol{\theta}))^2}{\sigma_i^2} \ .$$

  - The set of parameters $\boldsymbol{\theta}$ which maximize $\mathcal{L}$ is the same as those which minimize χ^2.

- Best-fit
  - N measurements X_i^{obs} at redshifts z_i with measurements errors σ_i, and the ϕCDM model predicts $X_i^{th}(\alpha, \Omega_{\rm m})$.
  - The χ^2 function is

$$\chi^2(\alpha, \Omega_{\rm m}) = \sum_i \frac{[X^{obs}(z_i) - X^{th}(z_i, \alpha, \Omega_{\rm m})]^2}{\sigma_i^2} \ .$$

  - Small values of χ^2 indicate a good fit. The parameters α^* and Ω^* that minimize χ^2 are called the best-fit parameters.

- There many observational data, the regular 3 ones are
  - Type Ia Supernovae [astro-ph.CO/0901.4804]
  - Baryon Acoustic Oscillation [astro-ph/0705.3323]
  - Cosmic Microwave Background [astro-ph.CO/1001.4538]
- From ϕCDM moedel, we have

$$H(\alpha, \Omega_{\rm m}) = H_0 \sqrt{\Omega_{\rm m} \left(\frac{a_0}{a}\right)^3 + \Omega_\phi(\alpha)} \text{ and } E(\alpha, \Omega_{\rm m}) \equiv \frac{H}{H_0}$$

here the parameter space is $(\alpha, \Omega_{\rm m})$.

- Confidence Level

Table 32.2: $\Delta\chi^2$ or $2\Delta \ln L$ corresponding to a coverage probability $1-\alpha$ in the large data sample limit, for joint estimation of m parameters.

$(1-\alpha)$ (%)	$m=1$	$m=2$	$m=3$
68.27	1.00	2.30	3.53
90.	2.71	4.61	6.25
95.	3.84	5.99	7.82
95.45	4.00	6.18	8.03
99.	6.63	9.21	11.34
99.73	9.00	11.83	14.16

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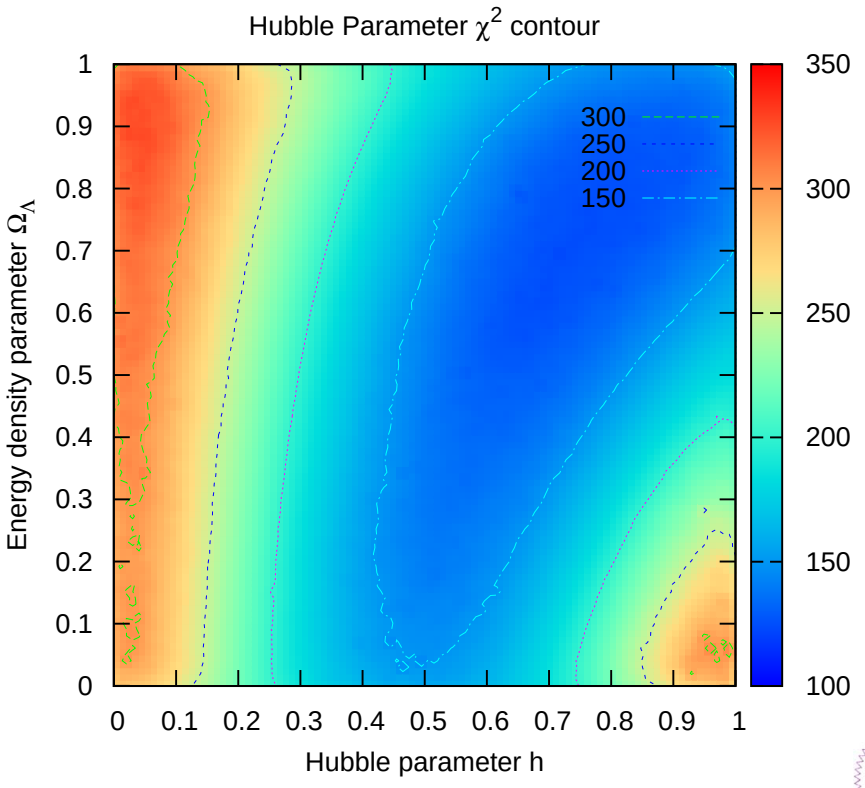
- Example
  - Use the Hubble parameter data to fit the ΛCDM model, $H = 100h\sqrt{\Omega_{\rm M}\left(\frac{a_0}{a}\right)^3 + \Omega_\Lambda}$ with $100h = H_0$, and the parameter space is (h, Ω_Λ).

z	$H(z)$	σ_H
0.09	69	12
0.17	83	8.3
0.27	70	14
0.4	87	17.4
0.88	117	23.4
1.3	168	13.4
1.43	177	14.2
1.53	140	14
1.75	202	40.4

[astro-ph.CO/0908.4597]



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- Homogeneity and isotropy of the Universe.
- For cosmic acceleration, the equation of state of dark energy $w_{\mathrm{DE}} < -\frac{1}{3}$.
- For ΛCDM model, it have "fine tuning" problem and "coincidence" problem.
- There are 3 alternative DE model: modified matter, modified gravity and inhohmogeneous Universe model.
- Quintessence is a slowly rolling scalar field candidate of the dark energy, and it require the condition $\dot{\phi}^2 < V(\phi)$ to realize the late-time cosmic acceleration.

- There two regular statistical methods, χ^2 and likelihood methods.
- The regular observational data we used are
  - Type Ia supernovae,
  - Baryon Acoustic Oscillation,
  - Cosmic Microwave Background.
- From statistical fitting result, we can give best-fit values and the constraints of the parameters in each model.

End

- Thanks for your attention!